

Automatic Detection of Oil Spills in ERS SAR Images

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IEEE Transactions on Geoscience and Remote Sensing, 1999

Abstract

This paper presents a framework for the automatic detection of oil spills in ERS Synthetic Aperture Radar (SAR) images. The proposed system consists of three main stages: detection of dark spots, feature extraction, and classification of detected regions as oil slicks or lookalikes. The classification approach combines statistical modeling with rule-based corrections and incorporates prior knowledge such as proximity to ships or oil platforms and environmental conditions like wind speed. Experimental results on 84 SAR images demonstrate high detection accuracy and strong discrimination between oil slicks and lookalikes.

1 Introduction

Automatic detection of marine oil spills in SAR imagery is an important task for environmental monitoring and pollution control. SAR sensors provide all-weather, day-and-night imaging capability, making them particularly suitable for operational oil spill surveillance.

However, oil spill detection is challenging because many natural oceanographic phenomena, referred to as lookalikes, also appear as dark regions in SAR images. Examples include low-wind areas, grease ice, current shear zones, rain cells, and internal waves. Even experienced human operators sometimes find it difficult to distinguish oil slicks from lookalikes.

The goal of this work is to develop a semiautomatic system that identifies dark regions with a high probability of being oil slicks and presents them to a human operator for final inspection, thereby significantly reducing manual workload.

2 SAR Imaging of Oil Spills and Lookalikes

Oil spills appear as dark regions in SAR images because oil dampens short surface waves responsible for Bragg scattering. The visibility and contrast of an oil slick depend on external conditions such as wind speed, wave height, and the amount and type of oil released.

The shape of an oil slick depends on whether the release originates from a stationary platform or a moving ship, as well as the wind and current history. Lookalikes often occur under low-wind conditions and may have shapes and intensities similar to oil slicks.

Important factors for discriminating oil slicks from lookalikes include proximity to known point sources such as ships or platforms, wind conditions, local contrast, and the homogeneity of the surrounding sea surface.

3 Dark Spot Detection

The detection of candidate oil spills is performed using adaptive thresholding. A local mean backscatter level is estimated within a large sliding window, and pixels with intensity values significantly below this mean are classified as dark.

Wind information is used to adjust the threshold level, since oil slick contrast decreases with increasing wind speed. Wind level is categorized into four classes ranging from low to high wind.

To improve detection accuracy in heterogeneous backgrounds, a clustering step is applied to each detected dark spot. If a detected region includes surrounding background pixels, the darkest cluster is retained as the candidate oil slick.

4 Slick Feature Extraction

For each detected dark spot, a feature vector is computed to characterize its shape, contrast, and surroundings. The extracted features include distance to the nearest bright object, number of detected spots in the scene, number of neighboring spots, homogeneity of surroundings, slick area, slick complexity, slick width, shape descriptors, local contrast ratio, border gradient, and power-to-mean ratio.

Some features describe prior knowledge, such as proximity to ships or oil platforms, while others quantify geometric and radiometric properties of the detected region.

These features form the basis for subsequent classification.

5 Slick Classification

The classification task aims to assign each detected dark spot to either the oil slick or lookalike class. The problem is context-dependent, requiring the incorporation of prior knowledge and environmental information.

The classifier combines three components:

- Prior probabilities reflecting the expected number of oil slicks and lookalikes in a scene.
- Class-conditional probability densities for the observed features.
- Rule-based corrections that account for known feature interactions.

Bayesian decision theory is used to compute posterior probabilities, and a loss function is applied to favor correct detection of oil slicks over rejection of lookalikes.

6 Statistical Modeling

Prior probabilities are defined based on the expected number of oil slicks in a scene and are adjusted using information about the presence of ships or oil platforms. The number of lookalikes is modeled using a geometric distribution whose parameter depends on wind level.

Class-conditional feature densities are modeled using multivariate Gaussian distributions. To handle data imbalance and limited training samples, a common diagonal covariance matrix is used for both oil slicks and lookalikes.

Feature densities are further conditioned on wind level and shape descriptors to improve modeling accuracy.

7 Rule-Based Density Corrections

Certain combinations of features strongly indicate either oil slicks or lookalikes. Since correlations between features are ignored in the Gaussian model, rule-based corrections

are applied to adjust the probability density ratios.

These rules account for slick area, number of detected regions, local contrast, slick width, and homogeneity of surroundings. The rules modify the statistical classifier output to reduce false alarms and improve robustness in complex scenes.

8 Methodology

The overall detection methodology consists of the following steps:

1. Acquire ERS SAR image and estimate wind level.
2. Detect dark spots using adaptive thresholding.
3. Refine detected regions using clustering.
4. Extract geometric, radiometric, and contextual features.
5. Compute prior probabilities based on scene context.
6. Estimate class-conditional probabilities using statistical models.
7. Apply rule-based corrections to probability densities.
8. Classify dark spots as oil slicks or lookalikes.

This integrated framework mimics the reasoning process of a trained human operator.

9 Experimental Results

The algorithm was evaluated on a dataset of 84 ERS SAR images selected by experienced operators. The dataset includes scenes with confirmed oil spills as well as scenes containing only lookalikes.

Oil slicks were categorized into thin linear slicks, wide regular slicks, wide irregular slicks, and piecewise linear slicks. A leave-one-image-out strategy was used to train and test the classifier.

Visual inspection of results shows successful detection of oil slicks in heterogeneous and complex ocean scenes.

10 Performance Analysis

Performance evaluation demonstrates that the proposed method correctly classified approximately 94% of oil slicks and 99% of lookalikes. Incorporating distance to bright objects and rule-based density corrections significantly reduced both missed detections and false alarms.

Comparative analysis of different classifier configurations confirms that the combination of statistical modeling and rule-based reasoning provides superior performance over purely statistical approaches.

These results indicate that the system is suitable for operational oil spill monitoring.

11 Discussion and Conclusion

This paper presents a robust semiautomatic system for oil spill detection in ERS SAR images. By combining prior knowledge, statistical feature modeling, and rule-based corrections, the method achieves high detection accuracy under diverse environmental conditions.

The framework reduces the need for manual inspection while retaining human oversight for final decision-making. The statistical formulation also provides a foundation for future improvements, including adaptive learning during operational use.